

RUBTSOVSK, Russian Federation

WMO: 360340

Lat: 51.576N

Lon: 81.199E

Elev: 719

StdP: 14.32

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-25.6	-37.0	0.7	-31.0	-31.0	1.0	-25.3	31.4	20.3	28.8	19.4	4.5	210	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	89.0	65.6	85.9	64.7	82.8	63.6	69.0	81.8	67.3	79.9	66.0	78.5	7.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	94.8	73.6	63.1	88.7	72.2	61.5	83.7	71.1	33.7	81.9	32.3	80.1	31.2	78.4	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.2	23.2	20.7	-34.9	95.1	9.0	3.3	-41.4	97.5	-46.6	99.4	-51.7	101.2	-58.2	103.6		
(5)			-35.1	71.9	8.9	2.4	-41.5	73.6	-46.7	75.0	-51.7	76.4	-58.2	78.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	1.8	5.9	21.0	42.3	56.4	66.9	69.2	65.4	54.0	40.9	23.2	9.0		
	DBStd	26.27	15.07	14.15	12.48	10.17	8.85	6.96	5.76	6.62	8.10	9.32	13.96	14.01		
	HDD50	6368	1494	1234	898	268	41	1	0	1	50	306	805	1270		
	HDD65	10191	1959	1654	1363	683	288	63	26	77	340	748	1255	1735		
	CDD50	2049	0	0	0	36	240	507	596	478	169	23	0	0		
	CDD65	398	0	0	0	0	22	120	157	90	9	0	0	0		
	CDH74	4192	0	0	0	16	343	1232	1398	1015	186	2	0	0		
	CDH80	1321	0	0	0	1	90	387	464	329	50	0	0	0		
(14) Wind	WSAvg	8.9	9.3	9.4	9.5	9.4	9.3	8.0	6.9	7.0	7.4	9.1	10.3	10.7		
(15) Precipitation	PrecAvg	13.4	0.7	0.7	0.8	0.8	1.3	1.3	2.2	1.4	1.1	1.2	1.1	0.8		
	PrecMax	22.4	3.1	2.3	2.8	2.3	2.9	2.9	5.9	3.2	2.8	3.0	2.3	1.8		
	PrecMin	6.9	0.0	0.1	0.3	0.0	0.0	0.1	0.3	0.3	0.1	0.3	0.1	0.1		
	PrecStd	3.1	0.6	0.6	0.6	0.5	0.8	0.9	1.6	0.8	0.6	0.6	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.8	35.9	53.9	76.2	88.8	91.1	93.5	92.9	86.2	71.8	54.1	35.9		
		MCWB	32.2	32.4	43.4	55.7	63.6	64.9	68.5	68.2	60.8	52.5	43.7	32.6		
	2%	DB	30.0	32.3	45.0	69.7	81.9	87.5	88.5	87.4	79.2	65.1	48.7	33.2		
		MCWB	27.6	29.6	38.8	52.5	59.4	64.3	67.3	65.2	58.7	50.1	40.6	30.7		
	5%	DB	26.6	29.4	40.2	64.8	77.8	84.4	85.1	83.6	74.3	60.1	44.0	30.5		
		MCWB	24.4	27.1	35.9	50.1	57.7	63.3	66.0	64.2	57.1	47.9	38.4	28.2		
	10%	DB	23.2	26.2	36.8	59.9	73.2	81.2	81.8	79.8	69.3	54.9	39.6	27.6		
		MCWB	21.3	24.0	33.6	47.4	55.9	62.7	65.3	62.7	54.8	45.5	35.7	25.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	33.2	44.0	56.9	64.9	68.5	72.6	70.5	63.0	53.9	45.6	33.6		
		MCDB	34.4	35.0	51.8	71.8	84.7	83.3	84.7	86.2	79.3	65.6	50.2	35.0		
	2%	WB	28.1	29.8	39.4	53.7	61.7	67.0	70.0	67.6	60.3	51.3	41.3	30.7		
		MCDB	29.7	31.8	44.5	66.9	76.5	81.2	82.3	81.8	75.2	63.1	46.4	32.8		
	5%	WB	24.7	27.2	36.1	50.9	59.4	65.6	68.3	65.8	58.3	48.9	38.5	28.2		
		MCDB	26.6	29.4	39.7	62.8	73.6	79.3	79.5	79.4	72.2	58.5	44.3	30.3		
	10%	WB	21.3	24.1	33.7	48.2	57.4	64.3	66.8	63.9	56.0	46.4	35.7	25.8		
		MCDB	23.1	26.2	36.6	59.4	70.7	77.6	78.0	76.7	67.5	54.3	39.6	27.6		
(35) Mean Daily Temperature Range	MDBR		15.7	18.1	17.0	20.7	22.5	21.1	19.5	21.9	21.8	17.2	13.5	14.9		
	5% DB	MCDBR	15.2	16.8	17.6	29.3	29.5	26.6	25.5	28.7	30.7	26.8	17.2	14.8		
		MCWBR	14.4	15.1	12.6	15.2	12.7	9.6	8.9	10.7	13.6	14.3	11.4	13.8		
	5% WB	MCDBR	15.2	17.1	16.3	27.3	24.8	22.6	20.6	24.1	27.5	24.6	15.2	15.4		
		MCWBR	14.8	16.0	12.0	15.0	12.2	9.3	8.4	10.3	13.2	14.0	10.8	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.338	0.389	0.413	0.415	0.413	0.434	0.410	0.370	0.356	0.342	0.308		
	taud		2.029	1.950	1.924	2.127	2.177	2.270	2.214	2.253	2.310	2.239	2.081	1.981		
	Ebn at Noon		210	235	247	260	266	267	259	259	257	232	191	180		
	Edh at Noon		26	38	48	44	44	41	42	39	32	29	25	23		
(44) All-Sky Solar Radiation	RadAvg		422	781	1239	1498	1915	2039	1846	1634	1208	659	347	292		
	RadStd		32	36	105	115	136	162	122	89	95	83	39	37		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)		N/A	N/A	N/A	N/A	-0.94	N/A	N/A	N/A	N/A

Nomenclature: See separate page